

BUSINESS UNDERSTANDING REPORT

Telco Customer Churn Analysis

Client: ABC Communications Ltd

Consulting Team: AnalystLab Africa Consulting

Role: Junior Data Analyst

Dataset: Telco Customer Churn Dataset

Tools: Python, Pandas, Matplotlib, Seaborn

1. Business Background

ABC Communications Ltd operates in the telecommunications industry, where maintaining existing customers is important for sustainable business performance. Customers have different service preferences, contract arrangements, payment methods, and levels of engagement with the company.

One of the major challenges faced by telecommunications companies is **customer churn**. Customer churn occurs when a customer terminates or stops using a company's services. High customer churn can result in lost recurring revenue, reduced customer lifetime value, and increased costs associated with acquiring new customers to replace those who leave.

Understanding the characteristics of customers who churn can help a company identify potential retention problems and develop more targeted strategies to retain valuable customers.

2. Business Problem

ABC Communications Ltd has engaged AnalystLab Africa Consulting to investigate customer churn and identify patterns that may help the company improve customer retention.

The company needs to understand **which types of customers are more likely to leave and what factors are associated with higher churn rates**. Without this information, management may have difficulty identifying high-risk customer segments and deciding where customer retention efforts should be concentrated.

The analysis will therefore examine customer characteristics, contract arrangements, tenure, services, and payment methods to identify patterns associated with customer churn.

3. Project Objective

The primary objective of this project is to analyze customer churn data and generate business insights that can support customer retention strategies at ABC Communications Ltd.

The analysis aims to:

- Understand the composition of the company's customer base.
 - Identify customer segments with higher churn rates.
 - Examine the relationship between contract type and customer retention.
 - Investigate whether customer tenure is associated with customer loyalty.
 - Identify services associated with higher or lower churn.
 - Determine which payment methods have higher churn rates.
 - Develop actionable recommendations that management can use to improve customer retention.
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4. Key Business Questions

The analysis will address the following business questions:

1. **What does the customer base look like?**
2. **Which customer segments have the highest churn?**
3. **Does contract type influence retention?**
4. **Does tenure affect loyalty?**
5. **Which services influence churn?**
6. **Which payment methods have higher churn?**
7. **What actions should management take?**

These questions provide the framework for exploring the dataset and connecting the analytical findings to business decisions.

5. Analytical Approach

The project will involve inspecting and preparing the dataset before conducting exploratory and business-focused analysis. The analysis will examine customer demographics, contract types, tenure, services, payment methods, charges, and churn status.

Python will be used to clean and analyze the data and create professional visualizations, including bar charts, pie charts, histograms, a box plot, and a correlation heatmap. The results will then be used to identify important customer churn patterns, business risks, opportunities, and potential retention strategies.

The analysis will focus on identifying **associations and patterns** in the data. These findings will support business recommendations but will not be interpreted as proof that a particular factor directly causes customer churn.

6. Expected Business Value

The analysis is expected to help ABC Communications Ltd better understand its customer base and identify groups that may require additional retention attention.

The findings can support management in developing targeted strategies such as improving early-stage customer engagement, reviewing contract options, investigating high-risk service segments, and evaluating payment processes.

By using data to identify customers and characteristics associated with higher churn, ABC Communications Ltd can make more informed retention decisions, potentially reduce avoidable customer losses, improve customer lifetime value, and strengthen long-term business performance.

7. Conclusion

Customer churn is an important business issue for telecommunications companies because losing existing customers can affect recurring revenue and increase the need for customer acquisition.

This project will use the Telco Customer Churn Dataset to investigate the characteristics and behaviors associated with customer churn at ABC Communications Ltd. The analysis will provide evidence-based insights into customer segments, contracts, tenure, services, and payment methods, which will form the basis for practical recommendations aimed at improving customer retention.